Fiscal Year:	FY 2022	**Task Last Updated:**	FY 11/19/2021
PI Name:	Macias, Brandon Ph.D.		
Project Title:	Investigating Structure and Function of the Eye		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	brandon.r.macias@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-483-2026
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Johnson Space Center Cardiovascular and Vision Laboratory		
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PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Became civil servant fall 2020; previously KBR/NASA Johnson Space Center. Prior to that until 2016, was at the University of California, San Diego.		
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	01/30/2019	**End Date:**	09/30/2033
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2033 per HHC element/JSC (Ed., 12/16/21) NOTE: End date changed to 1/24/2026 per HHC element/JSC (Ed., 4/8/21) NOTE: End date changed to 9/30/2025 per PI (Ed., 12/21/19)		
Key Personnel Changes/Previous PI:	December 2021 report: Added Dr. Millennia Young to support biostatistics with the retirement of Al Feiveson, who was removed from the list of project CoInvestigators during the last annual reporting period.		

COI Name (Institution):	Brunstetter, Tyson O.D., Ph.D. (NASA Johnson Space Center) Hargens, Alan Ph.D. (University of California, San Diego) Huang, Alex M.D., Ph.D. (Doheny Eye Institute) Karanjia, Rustum M.D., Ph.D. (Doheny Eye Institute) Laurie, Steven Ph.D. (KBR/NASA Johnson Space Center) Martin, Bryn Ph.D. (University of Idaho, Moscow) Sadda, Srinivas M.D. (Doheny Eye Institute) Smith, Scott Ph.D. (NASA Johnson Space Center) Zwart, Sara Ph.D. (University of Texas, Galveston) Lee, Stuart Ph.D. (KBR/NASA Johnson Space Center) Gibson, Charles O.D. (Coastal Eye Associates) Kramer, Larry M.D. (University of Texas Health Science Center, Houston) Greenwald, Scott Ph.D. (KBR/NASA Johnson Space Center) Jasien, Jessica Ph.D. (JesTech./NASA Johnson Space Center) Lytle, Jason Ph.D. (KBR/NASA Johnson Space Center) Marshall-Goebel, Karina Ph.D. (KBR/NASA Johnson Space Center) Pardon, Laura O.D., Ph.D. (KBR/NASA Johnson Space Center) Young, Millennia Ph.D. (NASA Johnson Space Center)
Grant/Contract No.:	Internal Project
Performance Goal No.:	
Performance Goal Text:	
Task Description:	This proposal will identify if ocular structure and function alterations occur at a greater frequency and magnitude during one-year missions compared to six-month and two-month expeditions and whether the recovery profile is dependent upon mission duration. In addition, this project will determine if changes in vascular structure and function are greater after one-year missions and if they contribute to alterations in ocular structure and function. The identification of structural and functional changes related to the development, progression, and recovery of Spaceflight Associated Neuro-ocular Syndrome (SANS) will provide NASA the information necessary to inform the risk posture for future interplanetary expeditions with duration of up to three years and to identify possible countermeasures. SPECIFIC AIMS: Conducting SANS research on two-month, six-month, and one-year crewmembers will enable us to objectively generate data to help NASA determine if and how SANS findings change with mission duration. Data from these missions will enable the development of a non-linear trend model that can be extrapolated to make predictions for spaceflights that are up to three years in duration. Therefore, the results of the "Investigating Structure and Function of the Eye" (iSAFE) study will help NASA to define the risk posture for future interplanetary expeditions and to identify possible countermeasures. Specific Aim 1: To determine if ocular structural changes develop to a greater degree (frequency or magnitude) during long-duration one-year spaceflight missions compared to findings during shorter length missions, and if recovery is prolonged after longer missions. Specific Aim 2: To determine if ocular vascular structure is altered to a greater degree during long-duration one-year spaceflight missions, and if recovery is more prolonged. Specific Aim 3: To determine if ocular function is altered to a greater degree during long duration one-year spaceflight missions, and the recovery profile. Specific Aim 4: To determine if measures of vascular structure and function are altered to a greater degree during long-duration one-year spaceflight missions, and if these vascular adaptations correlate with alterations in ocular structure and function. RELEVANCE AND MAP TO THE NASA HUMAN RESEARCH ROADMAP: This multi-project proposal is in response to NASA research announcement Human Exploration Research Opportunities (HERO), 80JSC017N0001-BPBA, Appendix C, Topic 1: Analyses of the Temporal Nature of Human Adaptation to Long-Duration Low-Earth Orbit Mission Virtual NASA Specialized Center of Research (VNSCOR). This proposal addresses multiple Human Research Program (HRP) Integrated Research Plan Gaps, including: • SANS101: Determine the relationship between fluid shifts (intravascular, interstitial, CSF) and ocular manifestations in astronauts during spaceflight. • SANS102: Determine the relationship between the fluid-shifts induced ocular changes and fluid shifts in the CNS, including whether elevated intracranial pressure or brain edema play a role.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Investigating Structure and Function of the Eye (iSAFE) research study will advance NASA's understanding of Spaceflight Associated Neuro-ocular Syndrome (SANS), an important human health and performance risk. This goal will be accomplished by quantifying how ocular alterations develop, progress, and recover as a function of spaceflight duration and by identifying underlying mechanisms. Results from this study are anticipated to lead to a temporal model of SANS progression during long-term missions and to inform the development of countermeasures. Given the unique environment of the International Space Station, commonly used ophthalmic instruments are being implemented in novel operational environments (e.g., electroretinography (ERG), optical coherence tomography (OCT) angiography, pneumotonometry). This work may lead to the adoption of these new hardware, software, or protocol elements in clinical practice, benefiting patients on Earth. SANS shares characteristics with several terrestrial ophthalmic diseases, such as papilledema, and iSAFE study results could provide new insights into mechanisms underlying these conditions.
	• The iSAFE Principal Investigator (PI) team has continued to work with the NASA Research Operations and Integration (ROI) Element to integrate this project into the Complement of Integrated Protocols for Human Research (CIPHER) studies. This effort included providing inputs to the Scientific Research Document (SRD); the iSAFE SRD is now baselined for all mission durations: Short, Standard, and Extended. • The Test Readiness Review (TRR) document supporting the implementation of the iSAFE Study activities in NASA Johnson Space Center Building 21 was submitted to ROI. • The iSAFE PI team supported the TRR for the Payloads Development Laboratory (PDL) in NASA Johnson Space Center Building 9, which is now approved. This TRR permits the use of the PDL for training crewmember operators to perform ERGs and pneumatonometry.

Task Progress:	• Supported the informed consent briefings (ICB) for 5 potential CIPHER subjects (03/04/2021, 04/29/2021 [x2], 11/15/2021, 12/01/2021) and developed an iSAFE overview video that has been integrated into the CIPHER ICB presentation. • Supported a hardware checkout of the ERG units designated for flight, flight backup, ground use by ROI, and for baseline data collection. • ERG and pneumatonometry flight hardware/units passed Science Verification Testing (SVT) and are now on the International Space Station. • Two iSAFE operators completed Telescience Center orientation in preparation for remote guidance activities, and remote guidance scripts were drafted for ERG and pneumatonometry. • Completed the Payloads Training Dry Run (PTDR) that was overseen by NASA Marshall Space Flight Center, certifying proficiency of Cardiovascular and Vision Laboratory (CVL) personnel (Co-Investigator Scott Greenwald) to train crewmember operators for ERG and pneumatonometry. • Two crewmember operators have completed the “Training” sessions for ERG and pneumatonometry (11/03/2021, 11/12/2021). • Preliminary dry runs of baseline data collection were completed in the NASA Cardiovascular and Vision Laboratory. • Steps have been taken to ensure readiness for implementation of inflight OCT angiography, which include obtaining the flight version of the OCT software and conducting preliminary remote guidance training for CVL personnel. • Preliminary training occurred for iSAFE operators on Dynamic Vessel data collection and analysis. • The plans for iSAFE were presented at the 2021 Human Research Program Investigators’ Workshop and at the JSC Chapter of the NASA Alumni League, both virtual. • Substantial progress has been made towards finalizing the iSAFE data analysis plan. • Substantial progress has been made towards finalizing the suite of standard operating procedures (SOPs) and operator logs.
Bibliography Type:	Description: (Last Updated: 06/03/2026)
Abstracts for Journals and Proceedings	Greenwald SH, Pardon LP, Jasien JV, Brunstetter T, Mercaldo ND, Hargens A, Huang A, Karanjia R, Laurie SS, Lee SMC, Marshall-Goebel K, Martin BA, Kramer LA, Sadda S, Smith SM, Zwart SR, Macias BR. "Investigating structure and function of the eye (iSAFE)." 2021 NASA Human Research Program Investigators’ Workshop, Virtual, February 1-4, 2021. Abstracts. 2021 NASA Human Research Program Investigators’ Workshop, Virtual, February 1-4, 2021. , Feb-2021
Abstracts for Journals and Proceedings	Greenwald SH. "How a headward fluid shift during weightlessness may cause spaceflight-associated neuro-ocular syndrome (SANS)." NASA Alumni League - JSC Chapter Meeting, Virtual, November 5, 2020. Program. NASA Alumni League - JSC Chapter Meeting, Virtual, November 5, 2020. , Nov-2020